Zheyu Zhuang

Researcher in Robot Learning and Physical AI

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Research Statement

How can a robot learn a new skill from a handful of demonstrations, and keep it working when the scene, the camera, or the robot itself changes? My research answers this by asking what a policy should perceive. I ground visual representations in action and in the robot's own body, and use structure such as symmetry and control theory so that limited data goes further. I test every idea on physical robots, which has given me a working understanding of the whole robot system, from sensors and calibration to real-time control. I lead this work in collaboration with PhD students in the group, as a postdoctoral researcher at KTH with Professor Danica Kragic, after a PhD in Robotics at the Australian National University with Professor Robert Mahony.

Research Directions

- **Action-grounded perception.** Which visual information matters for control? I learn it from action supervision and the robot's own body, so that policies stay reliable when appearance, viewpoint, spatial layout, or robot embodiment change.
- **Data-efficient imitation learning.** How far can a handful of demonstrations go? I use structure such as symmetry and policy-aware augmentation to get more out of each demonstration.
- **Learning with control-theoretic structure.** In my PhD, I combined deep learning with control theory, learning real-time closed-loop visual controllers structured by control-Lyapunov functions.
- **Integrated robot systems.** Complete systems that perceive, act, and work alongside people, from human-robot collaborative assembly to competition-winning warehouse picking.

Skills

Robotics systems
How the full stack fits together: sensors, multi-sensor synchronisation, calibration, communication, ROS integration, real-time control, and deployment on physical robots.

Model design and training
Design, train, and fine-tune visuomotor and vision-language-action (VLA) policies; imitation learning; representation learning; building on pretrained models such as DINO, SAM, and VLMs.

Simulation and evaluation
Benchmark models across simulation and hardware; design evaluation protocols; analyse failures and sim-to-real gaps.

Mentoring and collaboration
Guiding PhD, master's, and undergraduate researchers from idea to publication; tutoring; designing hands-on robotics workshops; setting an independent research agenda.

Multimodal data
Fuse RGB, depth, point clouds, event cameras, and proprioception; build data-collection pipelines; use augmentation and data generation to get more from scarce demonstrations.

Perception and control
3D perception, 6-DoF pose estimation, state estimation, visual servoing, and control-Lyapunov methods.

Tools: Python, PyTorch, ROS.

Research Experience

Postdoctoral Researcher, Robot Learning 2023 to Present

KTH Royal Institute of Technology, Stockholm, with Professor Danica Kragic

- Mentor and collaborate with PhD students from idea to publication.
- Build robot-learning systems end to end: sensor setup, synchronisation and calibration, ROS integra-

tion, real-time control, and policy deployment on physical robots.

- Design, train, and fine-tune visuomotor and VLA policies on multimodal inputs: RGB, depth, point clouds, and proprioception.
- Build data-collection and evaluation pipelines spanning simulation and real robots, and trace failures across the stack, from perception to control.
- Collaborate with the University of Freiburg and Aalto University.

PhD Researcher, Robotics 2017 to 2022

Australian National University, Australian Centre of Excellence for Robotic Vision, with Professor Robert Mahony

- Built a complete human-robot collaboration system, integrating action recognition, visual servoing, and handover on a real robot.
- Team ACRV member, winner of the 2017 Amazon Robotics Challenge with a custom Cartesian robot; responsible for hand-eye calibration.
- Contributed to 3D perception research: event-camera stereo, state estimation, and 6-DoF pose estimation.

Mentoring and Teaching

Research Mentoring 2017 to Present

ANU and KTH

Mentored PhD students on projects leading to publications at CoRL, RA-L, and ICRA, and supervised undergraduate research resulting in a publication at IROS.

Workshop Designer and Head Tutor 2020 and 2021

Robotic Vision Summer School

Designed a hands-on workshop series in which students integrated SLAM and learning-based object detection on real robots; parts were later incorporated into Intelligent Robotics (ECE4078) at Monash University.

Teaching Assistant 2017 to 2021

Australian National University

Tutored five graduate courses and one undergraduate course in systems engineering, robotics, and computer vision, running tutorials and laboratory sessions.

Education

PhD, Robotics 2017 to 2022

Australian National University
Australian Centre of Excellence for Robotic Vision

Supervisor: Professor Robert Mahony

BSc, Electronic and Communication Engineering; Mechatronics (Double Major) 2013 to 2016

Australian National University

First Class Honours

Publications

Project pages for recent papers are on my Personal Page.

Action-Grounded Perception

Zhuang, Z., Wang, R., Heppert, N., Hahn, J. F., Valada, A., Pokorny, F. T., and Kragic, D., "Attention from Action, for Action: Emergent Visual Bottlenecks for Policy Learning." *Conference on Robot Learning (CoRL), 2026.*

Wang, R., **Zhuang, Z.**, Kragic, D., and Pokorny, F. T., "Localized Visual Feature Aggregation via Focus Pooling for Visuomotor Policies." *Conference on Robot Learning (CoRL), 2026.*

Wang, R., **Zhuang, Z.**, Kragic, D., and Pokorny, F. T., "PALM: Enhanced Generalizability for Local Visuomotor Policies via Perception Alignment." *IEEE Robotics and Automation Letters (RA-L), 2026.*

Wang, R., **Zhuang, Z.**, Jin, S., Ingelhag, N., Kragic, D., and Pokorny, F. T., "Feature Extractor or Decision Maker: Rethinking the Role of Visual Encoders in Visuomotor Policies." *IEEE International Conference on Robotics and Automation (ICRA), 2025.*

Zhuang, Z., Kyrki, V., and Kragic, D., "Raising Body Ownership in End-to-End Visuomotor Policy Learning via Robot-Centric Pooling." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.*

Data-Efficient Imitation Learning

Zhuang, Z., Wang, R., Marchetti, G. L., Pokorny, F. T., and Kragic, D., "MirrorDuo: Reflection-Consistent Visuomotor Learning from Mirrored Demonstration Pairs." *Conference on Robot Learning (CoRL), 2025.*

Zhuang, Z., Wang, R., Ingelhag, N., Kyrki, V., and Kragic, D., "Enhancing Visual Domain Robustness in Behaviour Cloning via Saliency-Guided Augmentation." *Conference on Robot Learning (CoRL), 2024.*

Learning with Control-Theoretic Structure

Zhuang, Z., Yu, X., and Mahony, R., "End-to-End Multi-Instance Robotic Reaching from Monocular Vision." *IEEE International Conference on Robotics and Automation (ICRA), 2021.*

Zhuang, Z., Yu, X., and Mahony, R., "LyRN (Lyapunov Reaching Network): A Real-Time Closed Loop Approach from Monocular Vision." *IEEE International Conference on Robotics and Automation (ICRA), 2020.*

Zhuang, Z., Leitner, J., and Mahony, R., "Learning Real-Time Closed Loop Robotic Reaching from Monocular Vision by Exploiting a Control Lyapunov Function Structure." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019.*

3D Perception and State Estimation

Kennedy, G., Gao, J., **Zhuang, Z.**, Yu, X., and Mahony, R., "A General Approach to State Refinement." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021.*

Wang, Z., Pan, L., Ng, Y., **Zhuang, Z.**, and Mahony, R., "Stereo Hybrid Event-Frame (SHEF) Cameras for 3D Perception." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021.*

Yu, X., **Zhuang, Z.**, Koniusz, P., and Li, H., "6DoF Object Pose Estimation via Differentiable Proxy Voting Regularizer." *British Machine Vision Conference (BMVC), 2020.*

Integrated Robot Systems and Human-Robot Collaboration

Zhuang, Z., Ben-Shabat, Y., Zhang, J., Gould, S., and Mahony, R., "GoferBot: A Visual Guided Human-Robot Collaborative Assembly System." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.*

Morrison, D., Tow, A. W., McTaggart, M., …, **Zhuang, Z.**, Lehnert, C., Reid, I., Corke, P., and Leitner, J., "Cartman: The Low-Cost Cartesian Manipulator That Won the Amazon Robotics Challenge." *IEEE International Conference on Robotics and Automation (ICRA), 2018.*

Milan, A., Pham, T., Vijay, K., …, **Zhuang, Z.**, Lehnert, C., Lin, G., Reid, I., Corke, P., and Leitner, J., "Semantic Segmentation from Limited Training Data." *IEEE International Conference on Robotics and Automation (ICRA), 2018.*

Professional Service

Associate Editor 2025 to 2027

IEEE/RSJ IROS 2025 and 2026; IEEE ICRA 2027

Reviewer

IEEE RA-L, IEEE T-RO, RSS, CoRL, ICRA, and IROS

Awards

Best Team Project 2020

Australian Centre of Excellence for Robotic Vision Annual Symposium

Best Technical Demo 2019

Australian Centre of Excellence for Robotic Vision Annual Symposium

Amazon Robotics Challenge, First Place 2017

Team ACRV

References

Professor Danica Kragic
Postdoctoral Supervisor, KTH
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Professor Robert Mahony
PhD Supervisor, ANU
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Mr. Nick Heppert
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